Unit-2 :

MCQ & SHORT-Q/A

By : Dr. Snehal K Joshi

1. Which Python library is mainly used for data manipulation and analysis?
A) Matplotlib
B) Pandas
C) TensorFlow
D) Scikit-learn
✅ Correct Answer: B) Pandas

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2. NumPy is mainly used for which purpose in data analysis?
A) Data visualization
B) Database management
C) Numerical computing and array operations
D) Web development
✅ Correct Answer: C) Numerical computing and array operations

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3. Which of the following best defines regression?
A) A technique for clustering data
B) A method for classification
C) A statistical method to predict a continuous value
D) A method to reduce dimensionality
✅ Correct Answer: C) A statistical method to predict a continuous value

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4. In regression, the variable being predicted is called:
A) Independent variable
B) Dependent variable
C) Control variable
D) Random variable
✅ Correct Answer: B) Dependent variable

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5. Which term measures the relationship between two variables?
A) Variance
B) Standard Deviation
C) Correlation
D) Skewness
✅ Correct Answer: C) Correlation

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6. What is covariance used to measure?
A) Spread of a single variable
B) Relationship between two variables

C) Frequency of data
D) Outliers in data
✅ Correct Answer: B) Relationship between two variables

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7. Which of the following is a benefit of Machine Learning?
A) Manual data entry
B) Rule-based decision making
C) Automated prediction and decision making
D) Data storage only
✅ Correct Answer: C) Automated prediction and decision making

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8. Which step comes first in the Machine Learning life cycle?
A) Model training
B) Model evaluation
C) Data collection
D) Deployment
✅ Correct Answer: C) Data collection

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9. Which type of Machine Learning uses labeled data?
A) Reinforcement Learning
B) Unsupervised Learning
C) Supervised Learning
D) Semi-supervised Learning
✅ Correct Answer: C) Supervised Learning

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10. Which of the following is an application of Machine Learning?
A) Word processing
B) Image recognition
C) File compression
D) Data backup
✅ Correct Answer: B) Image recognition

Unit-2 :

MCQ & SHORT-Q/A

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11. Which Pandas function is used to display the first five rows of a dataset?
A) show()
B) preview()
C) head()
D) sample()
✅ **Correct Answer: C) head()**

12. Which NumPy function is used to calculate the mean of an array?
A) avg()
B) mean()
C) sum()
D) median()
✅ **Correct Answer: B) mean()**

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13. Which characteristic best describes a regression model?
A) It predicts categorical outcomes
B) It predicts continuous numeric values
C) It groups similar data
D) It reduces data dimensions
✅ **Correct Answer: B) It predicts continuous numeric values**

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14. Which variable is used as input in regression?
A) Dependent variable
B) Target variable
C) Independent variable
D) Output variable
✅ **Correct Answer: C) Independent variable**

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15. If correlation between two variables is zero, it means:
A) Strong relationship exists
B) No linear relationship exists
C) One variable controls the other
D) Variables are identical
✅ **Correct Answer: B) No linear relationship exists**

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16. Which of the following best describes Machine Learning?
A) Programming with fixed rules
B) Learning from past experience and data
C) Manual data processing
D) Database management
✅ **Correct Answer: B) Learning from past experience and data**

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17. Which step involves splitting data into training and testing sets?
A) Data visualization
B) Data preprocessing
C) Model deployment
D) Model monitoring
✅ **Correct Answer: B) Data preprocessing**

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18. Which algorithm is used in supervised learning?

A) K-means
B) Apriori
C) Linear Regression
D) DBSCAN
✅ Correct Answer: C) Linear Regression

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19. Which ML technique is used for grouping similar data points?
A) Classification
B) Regression
C) Clustering
D) Forecasting
✅ Correct Answer: C) Clustering

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20. Which real-world application uses Machine Learning?
A) Calculator
B) Email spam filtering
C) File storage
D) Text editor
✅ Correct Answer: B) Email spam filtering

Unit-2 :

MCQ & SHORT-Q/A

By : Dr. Snehal K Joshi

21. Which Pandas structure is used to store tabular data?
A) Array
B) Matrix
C) DataFrame
D) Dictionary
✅ Correct Answer: C) DataFrame

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22. Which NumPy data structure is most commonly used for numerical computation?
A) List
B) Tuple
C) Series
D) ndarray
✅ Correct Answer: D) ndarray

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23. In regression analysis, which of the following is a dependent variable?
A) Input variable
B) Output variable

C) Control variable
D) Random variable
✅ Correct Answer: B) Output variable

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24. Which of the following represents a strong positive correlation?
A) -0.9
B) -0.1
C) 0.0
D) +0.85
✅ Correct Answer: D) +0.85

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25. Which type of Machine Learning does not use labeled data?
A) Supervised Learning
B) Reinforcement Learning
C) Unsupervised Learning
D) Semi-supervised Learning
✅ Correct Answer: C) Unsupervised Learning

Unit-2 :

MCQ & SHORT-Q/A

By : Dr. Snehal K Joshi

26. Which Pandas function is used to get summary statistics of a dataset?
A) info()
B) describe()
C) summary()
D) stats()
✅ Correct Answer: B) describe()

---

27. Which NumPy function is used to generate an array of zeros?
A) zero()
B) zeros()
C) empty()
D) ones()
✅ Correct Answer: B) zeros()

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28. Which of the following is NOT a characteristic of regression?
A) It predicts continuous values
B) It establishes relationship between variables
C) It is used for clustering data
D) It is used for forecasting trends
✅ Correct Answer: C) It is used for clustering data

---

29. Which ML life-cycle step focuses on measuring model accuracy?
A) Data collection
B) Model training
C) Model evaluation
D) Deployment
✅ **Correct Answer: C) Model evaluation**

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30. Which of the following is an example of unsupervised learning?
A) Email spam detection
B) Stock price prediction
C) Customer segmentation
D) House price prediction
✅ **Correct Answer: C) Customer segmentation**

Unit-2 :

By : Dr. Snehal K Joshi

SHORT-QUESTION / ANSWERS

1. Explain the role of Pandas in automating Exploratory Data Analysis.

Answer:
Pandas is a Python library used for data manipulation and analysis.
It provides data structures like DataFrames to store tabular data efficiently.
Pandas simplifies tasks like data cleaning, filtering, aggregation, and summary statistics, which are essential in EDA.

---

2. What is NumPy and how is it useful in data analytics?
Answer:
NumPy is a Python library for numerical computation and array operations.
It provides fast and efficient methods to perform mathematical operations on large datasets.
NumPy is often used with Pandas for data preprocessing and calculations in analytics.

---

3. Define regression and explain its main purpose in data analysis.
Answer:
Regression is a statistical technique used to predict the value of a dependent variable based on one or more independent variables.
It helps identify relationships between variables and forecast trends.
Regression is widely used in business, finance, and scientific research for prediction and analysis.

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4. Explain the characteristics of a regression model.
Answer:
A regression model predicts continuous numerical values.

It assumes a relationship between dependent and independent variables.
Key characteristics include linearity, minimal errors, and generalizability to new data.

---

5. Differentiate between dependent and independent variables in regression.
Answer:
The dependent variable is the outcome we want to predict.
The independent variable(s) are the input features used to make the prediction.
For example, predicting sales (dependent) using advertising spend (independent).

---

6. What is covariance and how does it differ from correlation?
Answer:
Covariance measures how two variables vary together but does not indicate the strength of the relationship.
Correlation standardizes this measure, giving a value between -1 and +1.
Correlation is easier to interpret, while covariance is more raw and scale-dependent.

---

7. Explain the concept of Machine Learning.
Answer:
Machine Learning is a branch of artificial intelligence that enables systems to learn from data.
It allows computers to make predictions or decisions without explicit programming.
ML systems improve their performance over time as they process more data.

---

8. Describe the main benefits of Machine Learning.
Answer:
Machine Learning automates decision-making and prediction processes.
It can handle large datasets efficiently and identify hidden patterns.
It improves accuracy, reduces manual effort, and adapts to changing data.

---

9. Explain the steps involved in the Machine Learning life cycle.
Answer:
The ML life cycle starts with data collection and preprocessing.
Next, the model is trained, evaluated, and optimized.
Finally, the model is deployed and monitored for performance, with continuous learning applied as needed.

---

10. Differentiate between supervised and unsupervised learning with examples.
Answer:
Supervised learning uses labeled data to predict outcomes, such as predicting house prices.
Unsupervised learning finds patterns in unlabeled data, such as customer segmentation.
Supervised focuses on prediction, while unsupervised focuses on discovering structure in data.

---

11. What is the purpose of using Pandas describe() function in EDA?

Answer:
The describe() function provides summary statistics of numerical columns in a dataset.
It shows count, mean, standard deviation, minimum, maximum, and quartile values.
This helps in quickly understanding the distribution and spread of data.

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12. Explain the difference between linear and multiple regression.
Answer:
Linear regression predicts a dependent variable using a single independent variable.
Multiple regression uses two or more independent variables for prediction.
Multiple regression captures complex relationships between variables.

---

13. What is the importance of splitting data into training and testing sets in ML?
Answer:
Splitting data ensures the model is trained on one set and tested on another.
It helps evaluate the model's performance on unseen data.
This prevents overfitting and improves generalization.

---

14. Explain overfitting and underfitting in Machine Learning.
Answer:
Overfitting occurs when a model performs well on training data but poorly on test data.
Underfitting occurs when a model cannot capture underlying patterns in data.
Proper model selection and data preprocessing prevent both issues.

---

15. What is the difference between covariance and correlation with respect to scale?
Answer:
Covariance depends on the scale of variables and is not standardized.
Correlation is scale-independent and ranges from -1 to +1.
Correlation is easier to interpret and compare across datasets.

---

16. Name two Python libraries that help automate Exploratory Data Analysis and explain briefly.
Answer:
Pandas helps manipulate, clean, and summarize tabular data.
NumPy provides fast numerical computations and array operations.
Together, they simplify EDA by reducing manual calculations.

---

17. What is the difference between supervised and unsupervised learning?
Answer:
Supervised learning uses labeled data to predict outcomes (e.g., classification, regression).
Unsupervised learning works with unlabeled data to find patterns (e.g., clustering, dimensionality reduction).
Supervised focuses on prediction, unsupervised on discovering structure.

---

18. Explain one real-world application of supervised learning.

Answer:

Email spam detection is a real-world supervised learning application.
The model learns from labeled emails (spam or not spam) to classify new emails.
It helps automate filtering and improve user experience.

---

19. Explain one real-world application of unsupervised learning.

Answer:

Customer segmentation in marketing uses unsupervised learning.
The model groups customers based on purchase behavior or demographics without prior labels.
It helps businesses target promotions and improve customer engagement.

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20. Why is understanding independent and dependent variables important in regression?

Answer:

Identifying dependent and independent variables ensures correct model formulation.
It clarifies which variable to predict and which to use as input.
Proper identification improves model accuracy and interpretation.

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21. What is the importance of feature scaling in Machine Learning?

Answer:

Feature scaling standardizes input variables to a similar range.
It prevents variables with larger scales from dominating the model.
Techniques like normalization and standardization improve model convergence and performance.

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22. Explain the concept of residuals in regression analysis.

Answer:

Residuals are the differences between observed and predicted values in a regression model.
They measure how well the model fits the data.
Analyzing residuals helps detect patterns, outliers, and model accuracy.

---

23. How does Pandas help in handling missing data?

Answer:

Pandas provides functions like dropna() and fillna() to manage missing values.
It allows removing rows/columns with missing data or imputing values.
This ensures data quality and avoids errors during analysis or modeling.

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24. Explain the role of a correlation matrix in EDA.

Answer:

A correlation matrix shows pairwise correlation coefficients between variables.
It helps identify strong, weak, or no linear relationships.
It is useful for feature selection and understanding variable dependencies.

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25. What is the difference between R-squared and adjusted R-squared in regression?

Answer:

R-squared measures the proportion of variance explained by the model.
Adjusted R-squared accounts for the number of independent variables and penalizes unnecessary predictors.
Adjusted R-squared is more reliable when comparing models with multiple features.

---

26. How does unsupervised learning help in anomaly detection?

Answer:

Unsupervised learning identifies patterns without labeled data.
It can detect data points that deviate significantly from normal patterns.
Anomaly detection is used in fraud detection, network security, and quality control.

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27. Explain the concept of bias and variance in machine learning.

Answer:

Bias is the error due to overly simple assumptions in the model.
Variance is the error due to sensitivity to training data fluctuations.
Balancing bias and variance is essential to prevent underfitting or overfitting.

---

28. How do Pandas and NumPy complement each other in EDA?

Answer:

Pandas handles tabular data and provides convenient data manipulation tools.
NumPy efficiently performs numerical operations on arrays.
Together, they allow seamless data cleaning, analysis, and preparation for ML.

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29. Describe one limitation of regression models.

Answer:

Regression assumes a linear relationship between independent and dependent variables.
It may fail if the relationship is nonlinear or highly complex.
Regression is also sensitive to outliers, which can distort predictions.

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30. Explain the role of training and validation data in machine learning.

Answer:

Training data is used to build and optimize the model.
Validation data helps tune model parameters and evaluate performance before final testing.
This ensures the model generalizes well to unseen data.

QUESTION BANK OF UNIT- 2 :

LONG QUESTIONS / SHORT NOTES

[Refer Unit-2 notes and prepare answers from following]

1. Explain the role of Python libraries Pandas and NumPy in automating Exploratory Data Analysis. Provide examples of commonly used functions.

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2. Discuss regression analysis in detail. Explain its characteristics, assumptions, and applications in real-world scenarios.

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3. Explain the concepts of dependent and independent variables in regression. Provide examples from business or scientific datasets.

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4. Explain covariance and correlation. How are they calculated, and how are they useful in data analytics? Include examples.

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5. Explain the concept of Machine Learning in detail. Discuss how it differs from traditional programming and why it is important today.

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6. Describe the benefits of Machine Learning in business and industry. Provide examples of applications where ML has transformed operations.

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7. Explain the Machine Learning life cycle in detail. Discuss each stage including data collection, preprocessing, model training, evaluation, and deployment.

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8. Differentiate between supervised and unsupervised learning. Discuss their techniques, applications, and examples.

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9. Explain real-world applications of Machine Learning in various domains such as healthcare, finance, marketing, and IT. Provide detailed examples.

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10. Discuss the importance of Python-based EDA and Machine Learning in predictive analytics. Explain how data preprocessing, analysis, and ML together help in decision making.

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11. Explain the different types of regression techniques. Discuss linear, multiple, and polynomial regression with examples.

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12. Discuss the importance of feature selection in Machine Learning. How do Pandas and NumPy help in feature engineering?

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13. Explain in detail the role of exploratory data analysis (EDA) in the Machine Learning life cycle. Provide examples of how EDA improves model performance.

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14. Discuss the difference between covariance and correlation in statistical analysis. Explain with a practical dataset example.

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15. Explain the challenges faced while implementing Machine Learning projects in real-world scenarios and how they can be addressed.

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16. Describe supervised learning algorithms in detail. Discuss classification and regression algorithms with examples.

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17. Describe unsupervised learning algorithms in detail. Discuss clustering and dimensionality reduction techniques with examples.

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18. Explain how Python libraries like Pandas and NumPy are used for handling missing data, outliers, and data cleaning in EDA.

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19. Discuss the role of correlation in feature selection and predictive modeling. Explain positive, negative, and no correlation with examples.

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20. Explain the end-to-end workflow of a Machine Learning project, starting from data collection to deployment. Include tools, libraries, and techniques used at each stage.
